

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — TOPICAL WORKSHEET PACK: 15E.5

PRINCIPLES OF NMR SPECTROSCOPY

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	Topic 15: CARBOXYLIC ACID DERIVATIVES, SPECTROSCOPY AND CHROMATOGRAPHY
SUB-TOPIC:	15E.5: Principles of NMR Spectroscopy
TOTAL QUESTIONS:	2 Questions (Structured + MCQs + Calculations)
TOTAL MARKS:	5 Marks
EXAMINER FAQs:	1 Empirical Examiner FAQs & Traps Included

HOW TO USE THIS TOPICAL PACK:

1. Complete all structured questions under timed exam conditions.
2. Pay close attention to command words (Explain, Deduce, Calculate, State).
3. Review the complete worked mark scheme in Section C.
4. Study the Empirical Examiner FAQs in Section D to avoid common candidate traps.

SECTION A: TOPICAL PRACTICE QUESTIONS — 15E.5

Q1. 1. (a) Explain why tetramethylsilane, $\text{Si}(\text{CH}_3)_4$ (TMS), is added as an internal standard in NMR spectroscopy.

[3 Marks]

WCH14/01/Oct22/Q9(a)

Q2. (b) State why CDCl_3 or CCl_4 is used as a solvent in ^1H NMR rather than CHCl_3 or water.

[2 Marks]

WCH14/01/Oct22/Q9(b)

SECTION B: WORKED MARK SCHEME — 15E.5

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: 1. (a) Explain why tetramethylsilane, $\text{Si}(\text{CH}_3)_4$ (TMS), is added as an internal standard in NMR spectroscopy. 1. Gives a single sharp peak at $\delta = 0$ ppm. 2. Si is less electronegative than C, so 12 H atoms are highly shielded (far right of spectrum). 3. Non-toxic, volatile, and inert (easily recovered from sample).	[3]
Q2	Q2: (b) State why CDCl_3 or CCl_4 is used as a solvent in ^1H NMR rather than CHCl_3 or water. 1. Deuterated solvents (CDCl_3) contain deuterium (^2H) which does not absorb in the ^1H NMR frequency range. 2. Prevents solvent ^1H peak from obscuring sample peaks.	[2]

SECTION C: EMPIRICAL EXAMINER FAQs & COMMON TRAPS — 15E.5**FAQ #1: Why is TMS used as the standard at 0 ppm?****Category: NMR Standards****COMMON EXAMINER TRAP / MISTAKE:**

Stating TMS has 4 protons.

MODEL ANSWER / EXAMINER EXPECTATION:

TMS has 12 IDENTICAL protons (or 4 identical carbons), giving a single intense signal. Silicon's low electronegativity shields the nuclei, placing the peak at $\delta = 0.0$ ppm.